

Appendix B

Technical & Product — how the MVP works, written for a non-CS reader: scoring, retrieval, grounded generation, review, tests, and limitations.

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Appendix B — Technical & Product

CareerArth AI · Aryan Bhandari · for Professor Tushar Jaruhar

Written for a non-CS reader. What the machine actually does, why it was built that way, and where its edges are. Source: `careerarth-ai/` (code), `docs/01-08` (specifications).

B1. Architecture in one paragraph

A user fills in a structured web form. A small server then runs five steps in a fixed order: **validate** the input (including refusing anything without consent), **score** it with a fixed arithmetic rubric, **retrieve** the most relevant roles from a curated career library, **generate** a written analysis constrained to only use those inputs, **lint** that analysis to verify every claim cites its evidence — then saves the result and logs an event. A consultant can open any result in a review screen, edit or approve it, and the customer's page updates to "Consultant reviewed." (Figure 2 in the main report shows this visually.)

B2. ARTH scoring — deterministic by design

- 20 questionnaire items (5 per dimension), each answered 0–4 against anchored endpoints ("No idea" ↔ "Specific role + timeline").
- Each dimension = weighted item average scaled to 0–100, plus small fixed modifiers from profile facts (e.g., 5+ years in the same role: –10 Trajectory; no LinkedIn: –5 Human Capital). Modifiers clamp to [–15, +10]; scores clamp to [0, 100].
- Overall = $0.30 \cdot A + 0.25 \cdot R + 0.20 \cdot T + 0.25 \cdot H$, computed on unrounded values, rounded once. Bands: Strong 85–100 · Stable 70–84 · Watch 55–69 · Vulnerable 40–54 · Critical 0–39.
- All weights/modifiers/bands live in **one configuration file**; code reads it, tests verify it, documentation cites it — the three cannot silently drift apart.
- The weighted formula reproduces the production site's published sample scorecard (72/48/52/81 → 64). This is kept as a *regression test* — the honest label for calibrating to your own marketing example — and every score object carries a methodology note saying the rubric is heuristic and unvalidated against outcomes.

Why it matters commercially: same answers in, same score out, every time. The product's numbers are explainable to a customer and auditable by a skeptic — the AI is never the source of a number.

B3. Intake

An 8-screen branded wizard (~5–7 minutes): profile → goals → concern → four questionnaire screens → consent. Optional fields stay off the critical path. A demo dropdown can prefill any synthetic persona. Every screen transition logs an event, so drop-off *location* — not just rate — is measurable.

B4. Knowledge base and retrieval

- **18 occupation profiles** across technology, finance, and marketing/consulting, hand-curated from **O*NET** (the U.S. Department of Labor's public occupational database), each with summary, core skills, common transitions (with difficulty notes), outlook, volatility, and a source citation with retrieval date.
- Retrieval is deterministic keyword matching with field weighting and inverse-document-frequency damping (so "management" doesn't match everything). The user's current role and target role are always included when the library has them. Top 5 entries go to the report generator, each with human-readable match reasons.
- If a user's field isn't covered, the system says so on the report (a "coverage note") rather than bluffing — degraded gracefully, disclosed honestly.

B5. Report generation — grounded, structured, reviewed

The generator receives exactly three inputs: the validated profile, the fixed scores, and the retrieved library entries. Its rules: never state or adjust a number; tag every claim with its evidence (`profile:` / `score:` / `kb:` / `questionnaire:` references); use calibrated language ("likely," "based on what you've told us"); never promise outcomes; route ungroundable observations to "discuss with your consultant." Output is a fixed structure: diagnosis → dimension explanations → strengths → risks → tensions → 2–3 career paths with trade-offs → skill gaps → 90-day plan → open questions.

An automated **evidence lint** then checks every section: citations present, `kb:` references only to entries actually retrieved, score statements matching computed values. Drafts that fail don't proceed.

Two generation modes: with an API key, Claude (Opus) via structured JSON output; without one, a deterministic mock that fills the same structure from the data. **Every report produced during this project used the mock** — the Claude path is fully written but has never executed (no key was configured). Real AI report quality is therefore unobserved; estimated cost (\$0.05–\$0.15/report) comes from published prices × estimated token counts, not measurement.

B6. Consultant review

A dedicated screen shows the full report beside the profile, scores, and retrieval evidence. The consultant approves/flags/rejects per section, can edit the diagnosis and next-decision text inline, records confidence (1–5) and notes; minutes are tracked automatically and edits measured. Approval flips the customer badge to "Consultant reviewed." Exercised once in QA (6.5 minutes, by the developer, on a synthetic run) — a mechanism test, not evidence about real consultant workload.

B7. The six synthetic personas (QA fixtures)

Persona	Stresses	A/R/T/H → Overall (band)
p1 plateaued	Tenure penalty; the rubric's worked example	73/49/23/76 → 58 (Watch)
p2 early-career	Low network, high momentum pattern	69/73/75/38 → 63 (Watch)
p3 industry-switcher	Cross-industry retrieval, skill gaps	54/64/58/65 → 60 (Watch)
p4 gilded-stagnation	High pay ≠ high score; obsolete skills	60/34/10/55 → 42 (Vulnerable)
p5 high-performer	Upper bands; non-alarmist tone	94/80/100/81 → 88 (Strong)
p6 returner	Career break, sparse data, constraints	73/38/38/48 → 51 (Vulnerable)

Demonstrates: score differentiation (42–88), independent dimension movement, distinct retrieved path sets, clean sparse-data handling, lint passing on all six, review mechanics. Cannot demonstrate: demand, usefulness, WTP, accuracy in real careers, or real AI quality — these are QA fixtures authored by the developer.

B8. Tests — 42/42 passing (re-verified)

Suite	Tests	Covers
Scoring	16	Worked example, sample-score calibration, clamps, each modifier in isolation
Validation	8	Schema rejection paths, consent enforcement
Retrieval	10	Golden rankings, guaranteed inclusion, synonyms, unknown-industry fallback
End-to-end (mock)	8	Full pipeline, lint, persistence, events

No test touches the network or needs an API key — deliberate, so the whole suite is reproducible by anyone (including an examiner) with `npm test`.

B9. Privacy & ethics, as implemented

Consent enforced in code (pipeline refuses without it) · synthetic data flagged `isSynthetic` · name/email stripped from anything sent to the model · all data in local files, none committed · no fine-tuning · hedging and no-guarantee rules in both the prompt and the review checklist · heuristic-score disclaimer on the intake page, in every score object, and on the report footer.

B10. Honest limitations (technical)

1. Rubric weights heuristic and partly circular (calibrated to the marketing sample).
2. Self-report bias unmitigated in-product; human review is the only check.
3. 18 roles / 3 industries; outside that, nearest-match with a disclosed coverage note.
4. Keyword retrieval misses semantics the synonym map doesn't anticipate.
5. Mock reports are readable but formulaic; real-generation tone is unobserved.
6. Prototype infrastructure: single process, file storage, unauthenticated review URLs, no booking/payment/email. Fine for a demo and a pilot; unacceptable beyond.

B11. UI evidence (captured from the running MVP)

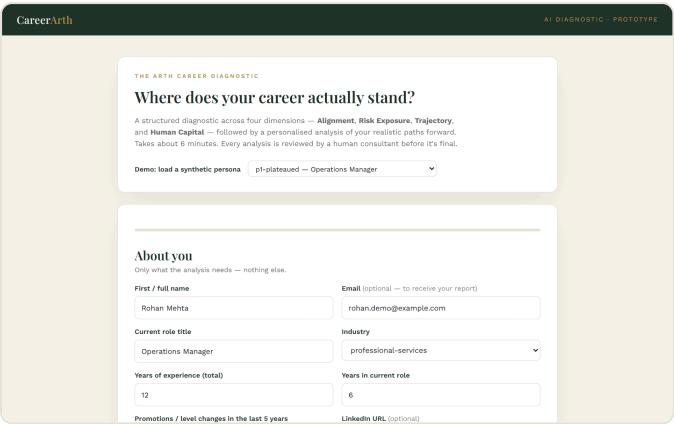


Figure B-1 — Intake step 1 with the p1 synthetic persona prefilled.

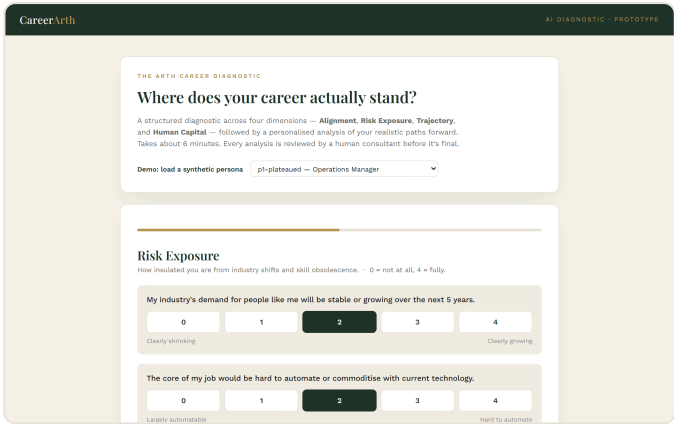


Figure B-2 — Risk Exposure questionnaire screen.

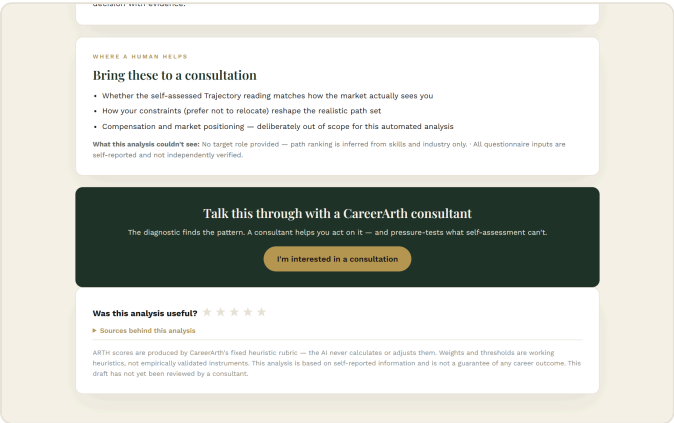


Figure B-3 — Consultation CTA and usefulness-rating widget — the funnel's conversion point.

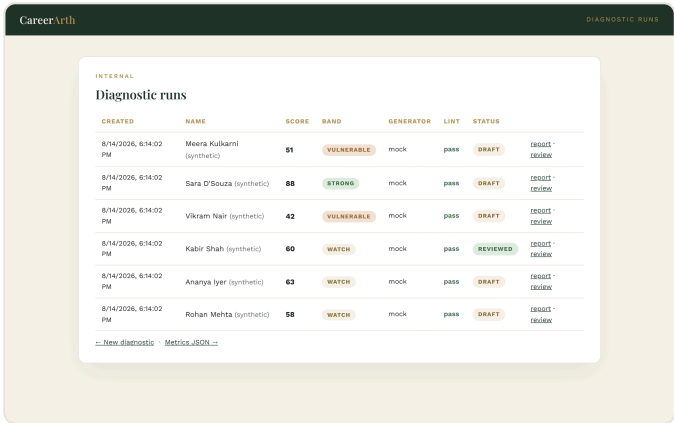


Figure B-4 — Runs index: all six synthetic personas, scores 42–88.